

```
//update source header parameters
nlh->nmsg_len = NMSG_SPACE(NLINK_MSG_SIZE);
```

```
//kernel id is 0
nlh->nmsg_pid = 0;
```

```
nlh->nmsg_flags = 0;
```

```
//copy the data
strcpy(NMSG_DATA(nlh), "Hello");
```

```
//update this if kernel belongs to
//multicast group
NETLINK_CB(skb).groups = 0;
//kernel id is 0
NETLINK_CB(skb).pid = 0;
```

```
//use this in case of unicast message
NETLINK_CB(skb).dst_pid = 2;
```

```
//update this in case of multicast
```

```
//message with multicast address
NETLINK_CB(skb).dst_groups = 0;
```

```
//unicast the message to process with
//process address 2
netlink_unicast(nlink, skb, 2, MSG_DONTWAIT);
```

```
//use this if multicast is to be done
//example is multicast address 3
//netlink_broadcast(nlink, skb, 0, 1<3, GFP_KERNEL);
```



References

- [1] <http://qos.itc.ku.edu/netlink/html/>
- [2] <http://linux.die.net/man/7/netlink>

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The abbreviations can either be local or global. Suppose you want to define a local abbreviation for 'international', you can type in 'intl' and use *C-x a i l* to define the expansion. It will prompt you with the message 'Mode expansion for "intl":'. You can then type the word 'international'. The next time you enter 'intl' followed by the space key, GNU Emacs will automatically expand the same for you. In order to define a global expansion, you need to use the *C-x a i g* command sequence. If you would like to remove all abbreviations for the current session, you can use *M-x kill-all-abbrevs* followed by the *Enter* key.

You can save the abbreviations to a file, say *~/emacs.d/.abbrev_defs*, using *M-x write-abbrev-file* for future use. You can also edit the stored abbreviations using the *M-x edit-abbrevs* shortcut. If you would like to view all the defined abbreviations, you can use *M-x list-abbrevs*. The relevant contents of *~/emacs.d/.abbrev_defs* are shown below:

```
;;-*-coding: utf-8;-*-
(define-abbrev-table 'Buffer-menu-mode-abbrev-table '())

(define-abbrev-table 'completion-list-mode-abbrev-table '())

(define-abbrev-table 'edit-abbrevs-mode-abbrev-table '())

(define-abbrev-table 'emacs-lisp-mode-abbrev-table
  '(
    ("intl" "international" nil 2)
```

```
))
```

```
(define-abbrev-table 'fundamental-mode-abbrev-table '())
```

```
(define-abbrev-table 'global-abbrev-table '())
```

```
(define-abbrev-table 'lisp-mode-abbrev-table '())
```

```
(define-abbrev-table 'sh-mode-abbrev-table '())
```

```
(define-abbrev-table 'shell-mode-abbrev-table '())
```

Printing

You can print the buffer contents from GNU Emacs using the *M-x print-buffer* command. If you would like to print a selected region, you can use the *M-x print-region* command. These will be printed with page numbers and headers. You can also print a buffer without page numbers using *M-x lpr-buffer*. Similarly, to print a region without page numbers, use *M-x lpr-region*. If you are already using the *Dired* mode, you can select the files that you want to print using the *P* shortcut, and then execute in *Dired* mode to print them.

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